

Milan Barot

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SUMMARY

Data Engineer with 6 years of experience specializing in **healthcare data engineering, payer/TPA systems, ETL, and legacy-to-modern platform migration**. Experienced with **HealthRules Payer (HRP), claims, eligibility, accumulator, provider, and financial data**, building and optimizing SQL-based data pipelines and reporting solutions across on-premises and cloud environments. Proven track record of translating complex healthcare business requirements into scalable data solutions while ensuring data accuracy, reliability, and production readiness. Currently expanding into **modern data engineering with Apache Spark, Databricks, and lakehouse architectures**, with hands-on experience building end-to-end data platforms using **Medallion architecture, Delta Lake, and AI-powered analytics**.

TECHNICAL SKILLS

Data Engineering	SQL, Python, PySpark, Apache Spark, SSIS, Azure Data Factory, ETL
Cloud & Lakehouse	Azure, Databricks, Delta Lake, Medallion Architecture
Databases/Data Warehouse	SQL Server, Azure SQL, PostgreSQL, DuckDB
BI & Reporting	Power BI, SSRS
Data Integration	EDI 834/837, REST APIs, FTP, ActiveBatch, SQL Server Agent
Workflow & DevOps	Airflow, Dagster, Docker, Git, Jenkins, Jira
Application & AI	FastAPI, React, Electron, ChromaDB, OpenAI API, Groq API

WORK EXPERIENCE

AmeriBen (UST Global | Implementation partner), remote (Health insurance)

SQL Server Developer/ Data Engineer January 2026 – Current

- Contributed to the migration of healthcare reporting from a legacy **GBAS** platform to an **HRP-based ODS** architecture, supporting the onboarding of self-funded employer groups and transition of **financial, utilization, claims, census, and clinical reporting** to the new platform.
- Developed and maintained **complex SQL Server reporting logic, stored procedures, merge scripts, and ETL workflows** supporting reports for **large claims, financial summaries, contract savings, utilization, lag analysis, and level-of-coverage census** across legacy and modernized data sources.
- Built **consolidated reporting datasets spanning both GBAS and ODS** during the platform transition, reconciling fundamentally different schemas and business structures to ensure consistent reporting for clients operating across legacy and HRP-derived data.
- Investigated data discrepancies through the **HPS → ODS transformation pipeline**, analyzing source HRP data, mappings, ETL logic, and downstream reporting objects; implemented corrective changes to ensure accurate representation of migrated healthcare data.
- Translated complex **payer/TPA business requirements** into production-grade SQL and reporting solutions, resolving ETL issues, SQL performance bottlenecks, cross-database dependencies, and data-quality issues across development, testing, and production environments.
- Collaborated with **business stakeholders and HRP teams** to understand healthcare reporting requirements, validate data mappings and business rules, and deliver technical solutions aligned with project requirements and migration milestones.

Tech Stack: SQL Server 2022, Azure SQL, SSRS, SSIS, ETL, HealthRules Payer (HRP)

Blue Cross Blue Shield Arizona (Mastek | Implementation partner), Phoenix, AZ (Health insurance)

Data Engineer Sept 2022 – Oct 2025

- Designed and implemented end-to-end **ETL pipelines** using **SSIS** and **SQL Server** to process **eligibility, claims, accumulator, and value-based care (VBC)** files, ensuring compliance with business requirements and vendor-specific standards. Collaborated with vendors such as **OptumRx, HealthEquity, Equality Health**, and others to deliver high-quality data solutions.
- Led offshore engineering resources across multiple ETL initiatives, managing task assignments, technical dependencies, and delivery timelines.
- Utilized **Azure Data Factory (ADF)** to design and orchestrate scalable **data integration pipelines** for migrating legacy member and claims data to **HealthRules Payer (HRP)** on **Azure Cloud**.

- Leveraged **Azure SQL Database** for staging, transformation, and persistent storage of high-volume healthcare data. Configured **ADF linked services, pipelines, and triggers** for automated, scheduled data ingestion from on-premises SQL Server to Azure.
- **Rebuilt critical ETL pipelines** using **SSIS** and **SQL Server** after the **CHC system hack (2024)**, restoring high-priority data workflows under tight deadlines with minimal downtime.
- Developed **complex data extracts** that combined on-prem warehouse data with select **cloud-based sources**, ensuring consistency, accuracy, and seamless integration across hybrid environments reducing execution time by 25 % on an average across the board.
- Collaborated closely with **Deloitte** on the HealthEquity project to develop **eligibility and claims files** and automated it end to end. Responsibilities included gathering business requirements, creating technical documentation, optimizing ETL pipelines, conducting performance testing, and ensuring smooth deployment into production environments.
- Automated data workflows by integrating **SQL Server Agent jobs** with **ActiveBatch**, setting up FTP connectivity with strict naming conventions and cadence-driven schedules for file delivery to vendors.
- Developed a consolidated **EDI framework for 834 and 837 enrollments, claims and dental**, using three core SSIS packages to automate the generation and delivery of over **35 distinct file types** to **CMS**. This architecture ensured secure data exchange and audit readiness while significantly reducing the manual maintenance and operational overhead previously required.
- Provided active production support for mission-critical **CMS data exchanges**, conducting over 25 successful production deployments and managing emergency releases maintaining compliance with federal reporting deadlines.
- Led the rapid resolution of critical production failures, ensuring minimal data loss during **high-volume enrollment** periods where direct connectivity with **CMS** was required
- Partnered with business stakeholders to gather requirements for new projects, conducting detailed assessments of complexity, technical effort, development timelines, and warranty periods. Worked with Mastek to draft **SOWs** for approval by Blue Cross Blue Shield, ensuring alignment with business objectives.
- Participated in Agile sprint meetings to track progress on deliverables, resolve blockers, and ensure timely completion of project milestones across multiple initiatives.

Tech Stack: Azure Data Factory (ADF), Azure SQL Database, C#, SQL Server, SSIS, SSRS, ActiveBatch, SQL Server Agent, FTP, EDI (834, 837), HealthRules Payer (HRP), Agile +4

CBRE, Dallas, TX (Commercial real estate services and investments)

Data Engineer Feb 2021 - Sept 2022

- Played a key role on the **Data Engineering & Reporting team** during the COVID-19 pandemic, delivering real-time data pipelines and analytics that supported CBRE's **Advisory business unit** across global commercial real estate portfolios.
- Designed, developed, and maintained robust **ETL pipelines** using **Python, SQL, and PostgreSQL** via CBRE's internal ETL framework (**Lymbic**). Translated complex business logic from real estate analysts into SQL/Python transformations for property usage, tenant risk, and leasing performance.
- Ingested diverse external data sources such as **Google Mobility Reports** and **BREAM energy efficiency assessments**, transforming them into structured datasets in **Parquet** and **CSV** formats. These outputs fed into **PowerBI dashboards** used by leadership to assess region-wise COVID impacts on commercial spaces.
- Applied data profiling and transformation using **Pandas and PySpark** for efficient in-memory processing of large CSV/Parquet files, enabling faster QA cycles and stakeholder feedback loops.
- Supported **Databricks** orchestration workflows — monitoring job execution, rerunning failed tasks, and assisting engineers with job recovery — ensuring daily and weekly batch pipelines remained consistent and reliable during volatile COVID data conditions.
- Worked in an **Agile delivery environment**, actively collaborating with analysts, product managers, and backend engineers in weekly sprint meetings to prioritize features, track bugs, and refine pipeline outputs based on stakeholder needs.
- Enabled high-impact real estate decisions by producing clean, trusted data for dashboards that revealed critical trends such as **building underutilization, shifting tenant behaviors, and geographic leasing risk**. These insights directly influenced CBRE's portfolio restructuring strategies and client advisory recommendations during the pandemic.

Tech Stack: Python, SQL, PostgreSQL, Power BI, Airflow, Pandas, Parquet, Agile

AmNet Services Inc., Fairfield, NJ (Telecommunications)

ETL/SQL Server Developer Oct 2019 - Aug 2020

- Developed SSIS packages to automate extraction and ingestion of network test data from Excel, flat files, and MS Access sources into a SQL Server staging environment.
- Designed and implemented a dimensional data warehouse model using **ERwin Data Modeler** to support analytical queries on network KPIs.

- Wrote complex **T-SQL stored procedures** for **data cleansing, aggregation, and validation** to ensure accuracy and consistency of telecom metrics.
- Created and maintained **SSRS reports** and **PowerBI dashboards** to visualize signal strength, call drop rates, and E911 test call success metrics.
- Automated **SQL Server Agent** jobs to schedule SSIS package executions and report generation, ensuring daily data processing without manual intervention.
- Collaborated with radio engineers, business analysts, and compliance teams in Agile sprints to gather requirements and iteratively improve data solutions.
- Conducted rigorous testing of ETL workflows and reports to guarantee data quality and system performance at scale.

Tech Stack: SSIS/SSRS, T-SQL, Power-BI, Agile

SELECTED PROJECTS

Kalshi-Prediction-Engine (AI-Powered Intelligence Lake-House) https://github.com/70milan/Kalshi_PredictionProject

- Architected a full-stack prediction market intelligence platform using a **Medallion (Bronze/Silver/Gold)** data lakehouse, processing 6 global news feeds, GDELT event streams, and Kalshi prediction market data through **PySpark** ETL with **Delta Lake ACID** guarantees.
- Engineered a **RAG**-powered inference engine that embeds Silver-layer data into **ChromaDB** (MiniLM-L6-v2), performs cascading time-windowed semantic search, and synthesizes bull/bear/verdict intelligence briefs via **Groq** (Llama 3.3 70B) — including automated "Ghost Pump" detection for unsupported price movements.
- Built a React Human-in-the-Loop trade terminal backed by a **FastAPI** bridge server, surfacing **AI-generated intelligence** briefs with Kelly Criterion bet-sizing and a Safe Mode guard for risk-controlled live trade execution on Kalshi via RSA-signed API calls.
- Deployed a 10-container **Docker** Compose production environment with a **custom ETL orchestrator** (**watermark-based** change detection, lock-file API coordination, once-daily settlement sweeps) running autonomously at \$0/month on local Windows infrastructure.
- Engineered a resilient, cost-efficient news ingestion pipeline across diverse political sources, leveraging RSS discovery, Trafilatura full-text extraction, and VADER sentiment analysis; optimized Kalshi series-first filtering to reduce API calls by approximately 98%.

Tech Stack: Python (PySpark/Pandas), GroqAPI DuckDB, ChromaDB, Delta Lake, Kalshi API, GDELT GKG, Sentiment Analysis (VADER/LLM), Docker, React.

RealTime_Context_Engine (AI-Powered Windows Runtime Host) https://github.com/70milan/RealTime_Context_Engine

- Developed a **dual-process desktop application** utilizing an **Electron** frontend and a **Python/FastAPI** backend to provide **real-time** interview guidance.
- Integrated **OpenAI GPT models** via **low-latency streaming (SSE)** to deliver generative feedback with a **<200ms TTFT** (Time To First Token).
- Built a custom **hardware-locked licensing system** using **RSA digital signatures** and **machine-specific HWIDs** to secure distribution.
- Engineered a high-performance **screen capture and audio processing pipeline** for real-time context analysis during live sessions.
- Automated a full **distribution pipeline** using **PyInstaller** and **Electron Builder**, incorporating **code signing** and an **automated update system**.

Tech Stack: Python (FastAPI/Uvicorn), JavaScript (Electron/Node.js), OpenAI API (GPT-4o/5), RSA Cryptography, SSE (Server-Sent Events), PyInstaller, Electron Builder.

EDUCATION

Bachelor of Science in Information Technology in Software Engineering March 2022

Washington University of Science & Technology, Vienna, VA

Bachelors in Applied Science in Computer Information Systems Dec 2019

The City University of New York, New York, New York

CERTIFICATIONS

Databricks Data Engineer Associates Oct 2026